Research Methodology

Module 1

Foundations of Research

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1 Meaning, Objectives, and Types of Research

1.1 What Is Research?

The word *research* comes from the old French *recercher*, meaning "to search again." In simple terms, research is a careful, organised search for new knowledge or a better understanding of something already known.

> **Research** Research is a **systematic**, **objective**, and **logical** process of collecting, recording, analysing, and interpreting data in order to answer a question or solve a problem and to add to the existing body of knowledge.

Different scholars phrase it differently, but the idea is the same:

- **Redman & Mory:** research is a "systematised effort to gain new knowledge."
- **Clifford Woody:** research involves defining problems, forming hypotheses, collecting and evaluating data, and reaching conclusions that either support or reject the hypothesis.
- **Kothari (C.R.):** research is the pursuit of truth with the help of study, observation, comparison, and experiment.

> **Note** Research is *not* just "looking things up" or copying facts from a book. Collecting information without analysing it or answering a question is **information gathering**, not research.

1.2 Objectives of Research

Every research study is carried out to achieve one or more of the following broad aims:

1. **Exploration**: to gain familiarity with a phenomenon or to achieve new insights into it (*exploratory* studies).
2. **Description**: to portray accurately the characteristics of an individual, group, or situation (*descriptive* studies).
3. **Frequency and association**: to find how often something occurs or how it is linked with something else (*diagnostic* studies).
4. **Testing hypotheses**: to test a hypothesis about a causal relationship between variables (*hypothesis-testing* studies).

Beyond these, researchers also pursue practical goals such as solving a real problem, improving a process, supporting a decision, or simply satisfying curiosity.

1.3 Types of Research

Research can be classified in several ways: by *purpose*, by *method*, and by *nature of data*. The syllabus focuses on four types.

Basic (Pure or Fundamental) Research

Basic research is carried out to **expand knowledge** and develop theory, without any immediate commercial or practical use in mind. It asks "why?" and "how does this work?"

- Driven by curiosity and the desire to understand.
- Results are generalisable and often published as theory.
- **Example:** studying how human memory works, or developing a new theorem about graph structures.

Applied Research

Applied research uses existing knowledge to **solve a specific, practical problem**. It is directed towards a particular need of an organisation, industry, or society.

- Driven by a practical problem and a usable outcome.
- Results are specific to a context, product, or population.
- **Example:** testing which user-interface layout reduces errors in a hospital billing system.

Exploratory Research

Exploratory research is conducted when a problem is **poorly understood** or little prior work exists. Its goal is not to give final answers but to *discover ideas, clarify the problem, and form hypotheses* for later study.

- Flexible, open-ended, and often qualitative.
- Typical methods: literature review, expert interviews, focus groups, case studies, pilot surveys.
- **Example:** informal interviews with students to find out why they avoid the college library.

Descriptive Research

Descriptive research aims to **describe** the characteristics of a population or phenomenon accurately. It answers *who, what, where, when, and how many*, but does not explain *why*.

- Structured, with clear questions decided in advance.
- Typical methods: surveys, observation, census, fact-finding studies.
- **Example:** a survey reporting the percentage of BCA students who own a laptop.

Comparison at a Glance

Type	Main Purpose	Key Question	Outcome
Basic	Build theory and knowledge	Why does it happen?	Theory, principles
Applied	Solve a practical problem	How can we fix it?	Solution, product, policy
Exploratory	Understand an unclear problem	What is going on?	Ideas, hypotheses
Descriptive	Describe characteristics	What is it like?	Facts, profiles, figures

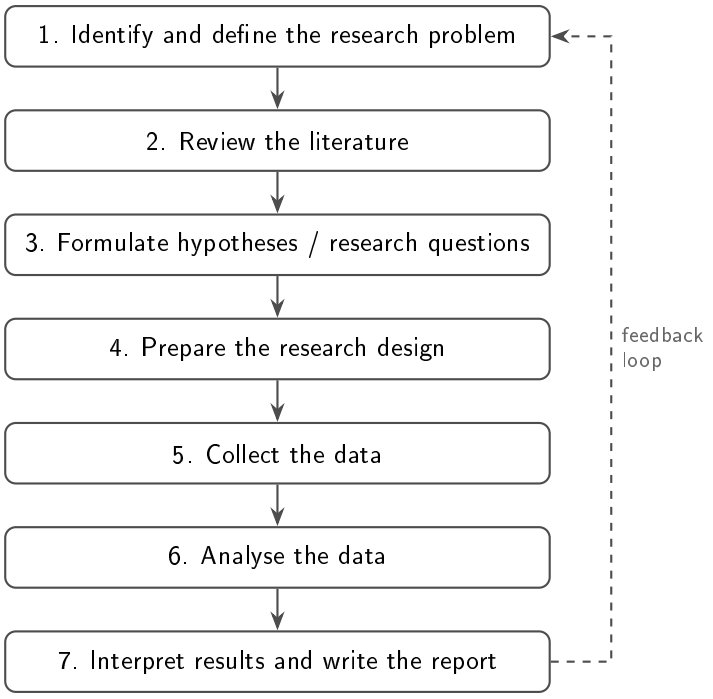
> **Exam Tip** A very common exam question is "Distinguish between basic and applied research." Write it as a short table with 4–5 points (purpose, motive, scope, outcome, example) and you will score full marks.

> **Note** Other classifications you may meet: **quantitative vs. qualitative** (numbers vs. words and meanings), **conceptual vs. empirical** (thought vs. observation), and **longitudinal vs. cross-sectional** (over time vs. one point in time).

2 Research Process and Characteristics of Good Research

2.1 The Research Process

The research process is the sequence of steps a researcher follows from the first idea to the final report. The steps overlap and often loop back, but the usual order is shown below.



Step-by-step explanation

1. **Identify and define the problem.** Decide what exactly you want to study and state it clearly.
2. **Review the literature.** Read earlier studies to see what is already known, what methods were used, and where the gaps are.
3. **Formulate hypotheses or research questions.** Turn the problem into testable statements or specific questions.
4. **Prepare the research design.** Plan how the study will be done: type of study, sample, tools, time, and cost.
5. **Collect the data.** Gather primary data (surveys, interviews, experiments, observation) or secondary data (books, records, databases).
6. **Analyse the data.** Organise, code, and test the data using suitable statistical or quali-

tative techniques.

7. **Interpret and report.** Explain what the results mean, link them to the objectives, draw conclusions, and present the findings.

> **Watch Out** The process is *not* strictly linear. Poor results in analysis may force you to redefine the problem or redesign the study. Treat the steps as a guide, not a rigid checklist.

2.2 Characteristics of Good Research

Good research is recognised by a set of qualities that make its findings trustworthy.

Characteristic	Meaning
Systematic	Follows a planned, step-by-step procedure rather than guesswork.
Logical	Uses sound reasoning (induction and deduction) to move from evidence to conclusion.
Empirical	Based on real observations and evidence, not on opinion or belief.
Objective	Free from the researcher's personal bias and feelings.
Replicable	Another researcher using the same method should obtain similar results.
Reliable and valid	Measures are consistent (reliable) and measure what they claim to (valid).
Clear purpose	Has a well-defined problem, objectives, and scope.
Ethical	Respects participants' consent, privacy, honesty, and credit to sources.
Generalisable	Findings can be applied beyond the sample studied, within limits.

> **Exam Tip** Remember the quick memory hook: **S-L-E-O-R**: Systematic, Logical, Empirical, Objective, Replicable. If you forget the rest, these five alone earn most of the marks.

3 Identifying and Defining a Research Problem

3.1 What Is a Research Problem?

A research problem is a **clear statement of an issue, gap, or difficulty** that the researcher wants to investigate and resolve. It is the starting point of the whole study. A badly chosen problem leads to a badly designed study, however good the later steps are.

3.2 Sources of Research Problems

- **Personal experience and observation** of everyday situations.
- **Literature:** gaps, contradictions, or "further research" suggestions in earlier papers.
- **Theory:** untested predictions or weak points in an existing theory.

- **Practical needs:** problems faced by industry, government, or society.
- **Discussion** with teachers, supervisors, and experts.
- **New technology or events** that create fresh questions.

3.3 Criteria for Selecting a Problem

Criterion	Ask yourself
Interest	Am I genuinely curious about this topic?
Relevance	Does it matter to the field, to practice, or to society?
Originality	Does it add something new, or fill a gap?
Feasibility	Do I have the time, money, data access, and skills?
Researchability	Can it be tested with data, or is it purely opinion?
Scope	Is it neither too broad nor too narrow?
Ethics	Can it be studied without harming anyone?

3.4 Steps in Defining the Problem

Defining a problem means turning a vague idea into a precise, workable statement. A well-known approach has these stages:

1. **Statement of the problem in a general way:** state the broad area and the central difficulty.
2. **Understand the nature of the problem:** discuss with those who have experience; know its origin and setting.
3. **Survey the available literature:** see what has been done, which theories apply, and what data exist.
4. **Develop ideas through discussion:** "brain-storm" with colleagues and experts to sharpen the thinking.
5. **Rephrase the problem into a working, specific form:** state it in analytical and operational terms so that it can actually be studied.

> **Note** **Technique:** Start broad and narrow down. For example, *"Online learning"* → *"Online learning among college students"* → *"Effect of daily screen time on exam performance of BCA students in West Bengal."*

> **Watch Out** Common mistakes: a topic that is *too broad* ("Study of social media"), a problem that is *too trivial* or already answered, a question that is *impossible to measure*, or one that needs *data you cannot access*.

4 Formulation of Research Questions and Objectives

4.1 Research Questions

A research question is the **specific question** the study sets out to answer. It converts the broad problem into something investigable.

Qualities of a good research question (often remembered as **FINER**):

F	**Feasible**	Can be answered with available time, data, and resources.
I	**Interesting**	Matters to the researcher and the audience.
N	**Novel**	Adds new knowledge or a new angle.
E	**Ethical**	Can be done without harm and with consent.
R	**Relevant**	Useful to the field, practice, or policy.

Examples of research questions on the topic "Smartphone use and study habits":

- What is the average daily smartphone use among BCA students? *(descriptive)*
- Is there a relationship between smartphone use and semester marks? *(relational)*
- Does limiting phone use improve concentration during study? *(causal)*

4.2 Research Objectives

Objectives state **what the study will achieve**. They are the research questions rewritten as action statements, and they guide the design, data collection, and analysis.

- A **general (broad) objective** states the overall aim.
- **Specific objectives** break the aim into smaller, measurable tasks.

> **Exam Tip** Begin each objective with an **action verb**: *to identify, to measure, to compare, to examine, to determine, to assess, to describe.* Avoid vague verbs such as *to know* or *to understand.*

Example. *General objective:* To study the effect of smartphone use on the academic performance of BCA students.

Specific objectives:

1. To measure the average daily smartphone use of BCA students.
2. To compare the semester marks of heavy users and light users.
3. To examine whether smartphone use is related to hours of study.

4.3 Hypotheses (Brief Link)

A **hypothesis** is a testable, tentative statement about the relationship between variables. Research questions lead to hypotheses when the study is going to test a relationship.

Null hypothesis (H_0)	There is *no* relationship or difference. Example: smartphone use has no effect on marks.
Alternative (H_1)	There *is* a relationship or difference. Example: smartphone use affects marks.

Note Hypothesis testing (t-test, chi-square, ANOVA) is covered in detail in a later module.

4.4 Questions vs. Objectives vs. Hypotheses

	Question	Objective	Hypothesis
Form	A question	An action statement	A testable statement
Asks / states	What do we want to find out?	What do we want to achieve?	What do we predict?
Example	Does phone use affect marks?	To examine the effect of phone use on marks.	Phone use lowers marks.

5 Variables in Research and Research Motivation

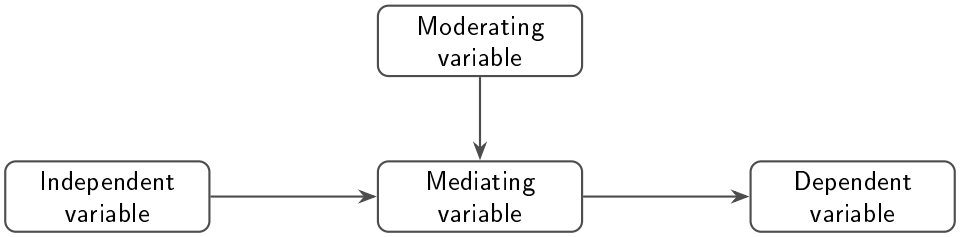
5.1 Variables in Research

Variable A variable is any **characteristic, quantity, or condition that can take different values** across people, objects, or situations. If something never changes, it is a *constant*, not a variable.

Examples: age, income, marks, gender, brand preference, temperature, hours of study.

Types of Variables by Role

Variable	Meaning	Example
Independent (IV)	The cause or input that the researcher changes or observes.	Hours of study
Dependent (DV)	The effect or outcome that is measured.	Exam marks
Control	Kept constant so it does not distort the result.	Same exam paper, same teacher
Extraneous	Outside factors that may influence the result unnoticed.	Student's sleep, mood
Confounding	An extraneous variable linked to both IV and DV, causing a false relationship.	Family income
Moderating	Changes the *strength or direction* of the IV–DV relationship.	Student motivation
Mediating (Intervening)	Explains *how or why* the IV affects the DV.	Better understanding



> **Exam Tip** To tell IV from DV, ask: *"Which one is affecting the other?"* The one doing the affecting is the **independent** variable; the one being affected is the **dependent** variable.

Worked example. *"Does regular exercise improve concentration in students?"*

- IV: amount of regular exercise.
- DV: concentration score.
- Control: age group, time of test, test difficulty.
- Extraneous: sleep quality, stress level.

Types of Variables by Nature of Measurement

Type	Meaning	Example
Continuous	Can take any value in a range, including decimals.	Height, weight, time
Discrete	Takes only separate, countable values.	Number of books owned
Categorical (Qualitative)	Placed into groups or categories.	Gender, city, department
Dichotomous	A categorical variable with only two values.	Yes/No, pass/fail

> **Note** Variables are also described by **levels of measurement**: *nominal* (names), *ordinal* (ranks), *interval* (equal gaps, no true zero), and *ratio* (equal gaps with a true zero). The level decides which statistical test you may use.

5.2 Research Motivation

Research motivation is the **reason or drive** that makes a person take up research. Understanding it helps in choosing a topic that one can sustain through a long, demanding process.

Common Motivations

Motivation	Description
Intellectual curiosity	Desire to know and understand something new.
Desire to solve problems	Wish to find practical solutions to real difficulties.
Academic requirement	Degree, thesis, project, or publication needs.
Career and recognition	Professional growth, prestige, or promotion.
Social service	Wish to improve society, policy, or public welfare.
Creativity and originality	Joy of creating something new or different.
Financial and institutional support	Grants, funding, or sponsorship from organisations.

Note Motivation can be **intrinsic** (comes from inside: curiosity, interest, satisfaction) or **extrinsic** (comes from outside: degree, money, recognition). Strong research usually has a healthy mix of both, but intrinsic motivation tends to last longer.

Watch Out Motivation must never override **honesty**. Pressure to publish or to get a result must not lead to faking data, selective reporting, or plagiarism.

6 Quick Revision Sheet

- **Research** = systematic, objective, logical search to answer a question and add knowledge.
- **Basic** builds theory; **Applied** solves a practical problem; **Exploratory** clarifies an unclear problem; **Descriptive** describes what exists.
- **Process:** problem → literature → hypothesis → design → data → analysis → report.
- **Good research:** systematic, logical, empirical, objective, replicable, ethical.
- **Defining a problem:** general statement → understand it → literature → discussion → rephrase specifically.
- **Good question (FINER):** feasible, interesting, novel, ethical, relevant.
- **Objectives** begin with action verbs and are measurable.
- **Variables:** IV causes, DV is affected; control is held constant; moderating changes strength; mediating explains why.
- **Motivation:** intrinsic (curiosity) and extrinsic (degree, recognition); keep it honest.

End of Module 1